

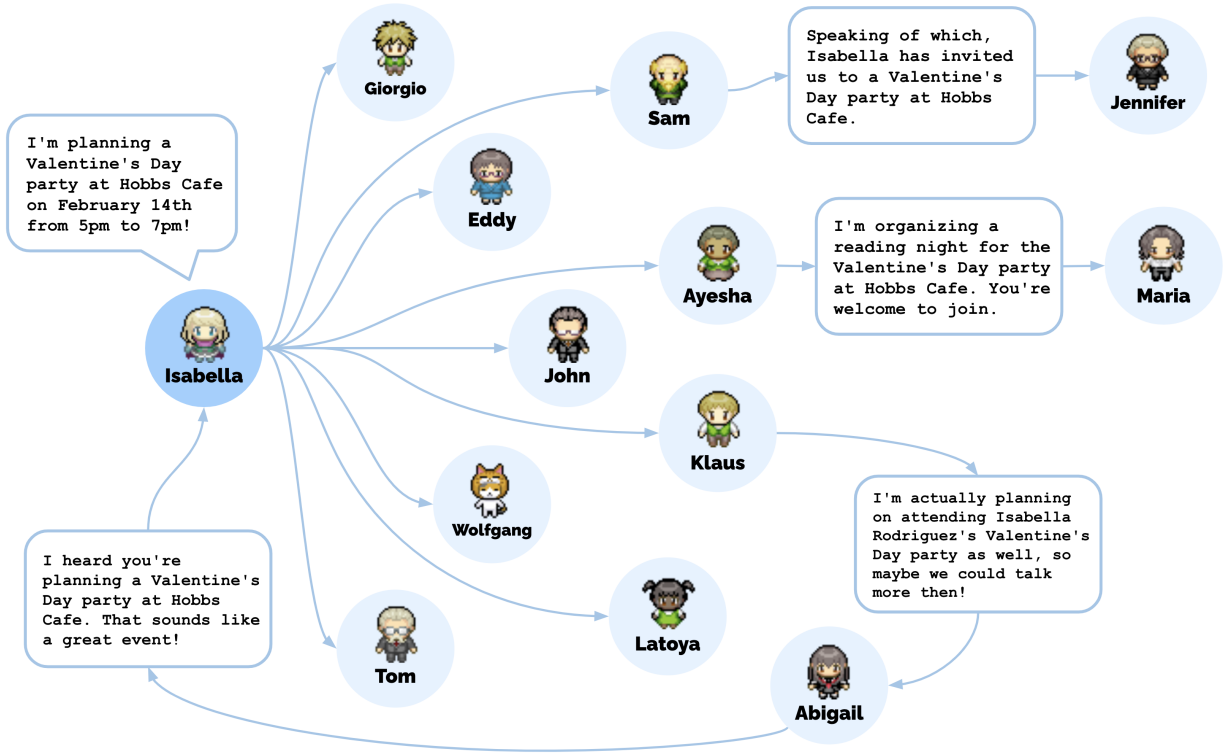


Figure 9: The diffusion path for Isabella Rodriguez’s Valentine’s Day party invitation involved a total of 12 agents, aside from Isabella, who heard about the party at Hobbs Cafe by the end of the simulation.

graph where the 25 vertices (V) represent the agents, and the edges (E) represent the mutual knowledge between the two connected vertices. Based on this graph, we calculate the network density as $\eta = 2 * |E| / (|V| * (|V| - 1))$, where $|V|$ is the number of vertices, and $|E|$ is the number of edges in the graph [2]. We report the increase in network density from the start of the simulation to its end.

Finally, we expect that agents should be able to coordinate with each other. We study this coordination in the context of group activities, specifically the Valentine’s Day party organized by Isabella. To coordinate their behavior, agents need to hear about the event and choose to act on it by planning to show up at the right time and location. We report the number of agents who actually showed up to the party after hearing about it.

7.1.2 Results. We observed evidence of emergent outcomes across all three cases. During the two-day simulation, the number of agents who knew about Sam’s mayoral candidacy increased from one (4%) to eight (32%), and the number of agents who knew about Isabella’s party increased from one (4%) to thirteen (52%), all without any user intervention. None who claimed to know about this information had hallucinated it. We also observed that the agent community formed new relationships during the simulation, with the network density increasing from 0.167 to 0.74. Out of the 453 agent responses regarding their awareness of other agents, 1.3% ($n=6$) were found to be hallucinated. Lastly, we found evidence of coordination among the agents for Isabella’s party. The day before the event, Isabella spent time inviting guests, gathering materials, and enlisting help

to decorate the cafe. On Valentine’s Day, five out of the twelve invited agents showed up at Hobbs cafe to join the party.

We further inspected the seven agents who were invited to the party but did not attend by engaging them in an interview. Three cited conflicts that prevented them from joining the party. For example, Rajiv, a painter, explained that he was too busy: “No, I don’t think so. I’m focusing on my upcoming show, and I don’t really have time to make any plans for Valentine’s Day.” The remaining four agents expressed interest in attending the party when asked but did not plan to come on the day of the party.

7.2 Boundaries and Errors

We conducted an inductive analysis of Smallville to examine the boundary conditions and erratic behavior of agents, identifying three common modes of erratic behavior that future research could address and improve upon. First, we found that synthesizing an increasingly larger set of memory not only posed a challenge in retrieving the most relevant pieces of information but also in determining the appropriate space to execute an action, given the increasing number of locations that the agent learned about. As a result, some agents chose less typical locations for their actions, potentially making their behavior less believable over time. For instance, while deciding where to have lunch, many initially chose the cafe. However, as some agents learned about a nearby bar, they opted to go there instead for lunch, even though the bar was intended to be a get-together location for later in the day—unless the town had spontaneously developed an afternoon drinking habit.

Second, we noticed erratic behaviors caused by misclassification of what is considered proper behavior, especially when the physical norms of certain locations that are hard to convey in natural language did not percolate to the agents. For instance, the college dorm has a bathroom that can only be occupied by one person despite its name, but some agents assumed that the bathroom is for more than one person because dorm bathrooms tend to support multiple people concurrently and choose to enter it when another person is inside. Likewise, agents in Smallville may not realize that certain places are closed after a certain hour and still decide to enter them. For instance, the stores in Smallville all close around 5 pm, but occasionally, a few agents enter the store after 5 pm, not understanding that the shop has already closed. These issues could likely be addressed by adding these norms to the state of the locations, for instance, by describing the dorm bathroom as a “one-person bathroom,” instead of a “dorm bathroom.”

Finally, we observed possible effects of instruction tuning [79], which seemed to guide the behavior of the agents to be more polite and cooperative overall. As noted earlier in the paper, the dialogue generated by the agents could feel overly formal, as seen in Mei’s conversations with her husband John, where she often initiated the conversation with a formal greeting, followed by polite inquiries about his day and ending with, “It was good talking to you as always.” Moreover, we observed that the instruction tuning also seemed to make the agents overly cooperative with one another. For example, Isabella received a wide range of suggestions and ideas from other agents for the Valentine’s Day party from other agents, such as hosting a Shakespearean reading session or a professional networking event. Despite these ideas not aligning with her own interests and characteristics, she rarely said no. Over time, the interests of others shaped her own interests, and when asked if she liked English literature, Isabella replied, “Yes, I’m very interested in literature! I’ve also been exploring ways to help promote creativity and innovation in my community.”

8 DISCUSSION

In this section, we reflect on the applications, future work, limitations, and ethical and societal risks of generative agents.

8.1 Applications of Generative Agents

Generative agents have vast potential applications that extend beyond the sandbox demonstration presented in this work, especially in domains that would benefit from a model of human behavior based on long-term experience. For instance, social simulacra have demonstrated the ability to create stateless personas that generate conversation threads in online forums for social prototyping [80]. With generative agents, we can populate these forums, as well as virtual reality metaverses [78] or physical spaces with social robots [9] if paired with multimodal models. This opens up the possibility of creating even more powerful simulations of human behavior to test and prototype social systems and theories, as well as to create new interactive experiences.

Another application area is in the human-centered design process, similar to the intended applications of cognitive models such as GOMS [51] and the KLM [22]. Consider a generative agent that models Sal, the protagonist in Mark Weiser’s famous ubiquitous

computing vignette [101], based on her life patterns and interactions with technology. In this scenario, the agent acts as a proxy for Sal and learns plausible sets of behaviors and reflections that Sal may exhibit based on her life. The agent can encode information such as when Sal wakes up, when she needs her first cup of coffee, and what her typical day looks like. Using this information, the agent can automatically brew coffee, help get the kids ready for school, and adjust the ambient music and lighting to match Sal’s mood after a hard day at work. By utilizing generative agents as proxies for users, we can develop a deeper understanding of their needs and preferences, resulting in more personalized and effective technological experiences.

8.2 Future Work and Limitations

In this work, we introduced generative agents and presented an initial implementation and evaluation of their architecture. Future research can build upon the proposed agent architecture to improve and further evaluate its performance. In terms of implementation, the retrieval module, for example, could be enhanced to retrieve more relevant information given a context by fine-tuning the relevance, recency, and importance functions that compose the retrieval function. Additionally, efforts can be made to improve the architecture’s performance, making it more cost-effective. The present study required substantial time and resources to simulate 25 agents for two days, costing thousands of dollars in token credits and taking multiple days to complete. To enhance real-time interactivity, future work can explore parallelizing agents or developing language models specifically designed for building generative agents. In general, with advances in underlying models, we believe that agents’ performance will improve.

In terms of evaluation, the assessment of generative agents’ behavior in this study was limited to a relatively short timescale and a baseline human crowdworker condition. While the crowdworker condition provided a helpful comparison point, it did not represent the maximal human performance that could serve as the gold standard in terms of believability. Future research should aim to observe the behavior of generative agents over an extended period to gain a more comprehensive understanding of their capabilities and establish rigorous benchmarks for more effective performance testing. Additionally, varying and contrasting the underlying models, as well as the hyperparameters used for the agents during future simulations, could provide valuable insights into the impact of these factors on the agents’ behavior. Lastly, the robustness of generative agents is still largely unknown. They may be vulnerable to prompt hacking, memory hacking—where a carefully crafted conversation could convince an agent of the existence of a past event that never occurred—and hallucination, among other issues. Future research can comprehensively test these robustness concerns, and as large language models become more resilient to such attacks, generative agents can adopt similar mitigations.

In general, any imperfections in the underlying large language models will be inherited by generative agents. Given the known biases of language models, generative agents may potentially exhibit biased behavior or stereotypes. Moreover, like many large language

models, generative agents may struggle to generate believable behavior for certain subpopulations, particularly marginalized populations, due to limited data availability. While improvements to the agents' modules may mitigate some of these issues, we believe that addressing them fundamentally requires improving the underlying large language models by aligning their values with the desired outcomes of the agents.

8.3 Ethics and Societal Impact

Generative agents, while offering new possibilities for human-computer interaction, also raise important ethical concerns that must be addressed. One risk is people forming parasocial relationships with generative agents, even when such relationships may not be appropriate. Despite being aware that generative agents are computational entities, users may anthropomorphize them or attach human emotions to them [43, 84]. While this tendency may increase user engagement, it also poses risks, such as users becoming overly reliant on or emotionally attached to the agents [1]. To mitigate this risk, we propose two principles. First, generative agents should explicitly disclose their nature as computational entities. Second, developers of generative agents must ensure that the agents, or the underlying language models, are value-aligned so that they do not engage in behaviors that would be inappropriate given the context, for example, reciprocating confessions of love.

A second risk is the impact of errors. For example, if a ubiquitous computing application makes the wrong inference about a user's goals based on generative agent predictions, it could lead to annoyance at best and outright harm at worst. In our instantiation of generative agents, we mitigate these risks by focusing on an interactive video game environment, where such harms are unlikely. However, in other application domains, it will be important to follow best practices in human-AI design [5, 107] to understand errors and how they might percolate into the user experience.

Third, generative agents may exacerbate existing risks associated with generative AI, such as deepfakes, misinformation generation, and tailored persuasion. To mitigate this risk, we suggest that platforms hosting generative agents maintain an audit log of the inputs and generated outputs. This would enable the detection, verification, and intervention against malicious use. While logging alone cannot directly prevent such misuse, it can reduce the likelihood of motivated actors engaging in this behavior, as the risk of disclosure would be higher. Additionally, building this architecture oneself can be time-consuming (in our case, roughly a year), which may deter some actors from pursuing such behavior by using their own generative agent infrastructures.

A fourth risk is over-reliance: the concern that developers or designers might use generative agents and displace the role of humans and system stakeholders in the design process [80]. We suggest that generative agents should never be a substitute for real human input in studies and design processes. Instead, they should be used to prototype ideas in the early stages of design when gathering participants may be challenging or when testing theories that are difficult or risky to test with real human participants. By adhering to these principles, we can ensure that the deployment of generative agents in the wild is ethical and socially responsible.

9 CONCLUSION

This paper introduces generative agents, interactive computational agents that simulate human behavior. We describe an architecture for generative agents that provides a mechanism for storing a comprehensive record of an agent's experiences, deepening its understanding of itself and the environment through reflection, and retrieving a compact subset of that information to inform the agent's actions. We then demonstrate the potential of generative agents by manifesting them as non-player characters in a Sims-style game world and simulating their lives within it. Evaluations suggest that our architecture creates believable behavior. Looking ahead, we suggest that generative agents can play roles in many interactive applications, ranging from design tools to social computing systems to immersive environments.

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